MTH104

Quiz no 1

2024

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The set R of real numbers is complete.

True

false

True.

What is the Cartesian product of the set {1, 2, 3} with itself?

A×A={(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)}

The set Q of rational numbers is complete.

True

false

False.

Given the set A={0,1,2} with the relations R={(0,1), (1,0)} and S={(0,0), (2,1), (1,0)}. Then S ∘ R = ____________________.Select the correct option

S ∘ R={(0,0), (0,1), (2,2)}

S ∘ R={(0,1), (1,1), (2,0)}

S∘R={(0,0),(0,1),(2,0)}

Let a and b be real numbers. Then |a ± b| ≥ ||a|| - ||b||.

True

False

The statement is false.

What is the Cartesian product of the empty set with any other set?

Select the correct option

It results in the other set

None of these

It results in a non-empty set

It results in an empty set

"It results in an empty set."

Consider two relations R and S. Then for the composition of relation R o S, the following is true:Select the correct option

R o S = S o R

R o S ≠ S o R4

Greatest common divisor of 88 and 135 is _________.Select the correct option

38

81

1

35

Common divisors: 1

So, the greatest common divisor of 88 and 135 is 1. Therefore, the correct option is not listed, it is 1.

Let A = {1, 2, 4}, and let R be a relation on A defined by "x divides y", written x|y. Then _________________.Select the correct option

R = { (1, 2), (1, 4), (2, 2), (2, 4), (4, 4)}

None

R = {(1, 1), (1, 2), (1, 4), (2, 4)}

R = {(1, 1), (1, 2), (1, 4), (2, 2), (2, 4), (4, 4)}

R={(1,2),(1,4),(2,2),(2,4),(4,4) –

R={(1,2),(1,4),(2,2),(2,4),(4,4)}

Prime factorization of 90 is __________.Select the correct option

none of these

3 × 3 × 5 × 2

3 × 1

5 × 2

9 × 5 × 2

2×3×3×5

Least common multiple of 9 and 12 is _________.Select the correct option

12

36

92

4

The correct option is: 36

The interval satisfying the inequality 0 ≤ 2x ≤ 3 is __________.Select the correct option

(0, 1/3]

None

[0,1/3]

(0,1/3)

The correct option is: [0,1/ 3].

Which of the following set is unbounded?Select the correct option

none

A = {x : |x| < 4}

B = {x : |x| ≤ 3}

C = {1,2,3,...}

So, the correct option is: none.

Findxandygiven(3x,x−y)=(3,3).

Select the correct option

(1, -2)

(1, 2)

(-1, -2)

(-1, 2)

(1,−2).

Given A={a,b} , then A × A = ____________________.Select the correct option

{(a,a), (a,b), (b,b)}

{(a,a), (a,b), (b,a), (b,b)}

{(a,a), (b,a), (b,b)}

{(a,a), (a,b), (b,a)}

{(a,a),(a,b),(b,a),(b,b)}.

Given A={1,2} and B={5,6,7}, then A × B= ____________________.Select the correct option

{(1,5), (1,6), (1,7), (2,5), (2,6)}

{(5,1), (5,2), (6,1), (6,2), (7,1), (7,2)}

{(5,1), (5,2), (6,1), (6,2), (7,1)}

{(1,5), (1,6), (1,7), (2,5), (2,6), (2,7)}

A×B={(1,5),(1,6),(1,7),(2,5),(2,6),(2,7)}

{(1,5),(1,6),(1,7),(2,5),(2,6),(2,7)}.

What is the Cartesian product of the set {1, 2, 3} with itself?Select the correct option

{(1, 1, 1), (2, 2, 2), (3, 3, 3)}

{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)}

{(1, 1), (2, 2), (3, 3)}

{(1, 2, 3)}

A×A={(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)}

The distance between -2 and 14 on the real line is __________.Select the correct option

15

16

14

-16

Therefore, the correct option is: 16.

Given A={1,2} and B={5,6,7}, then B × A = ______________________.Select the correct option

{(1,5), (1,6), (1,7), (2,5), (2,6), (2,7)}

{(1,5), (1,6), (1,7), (2,5), (2,6)}

{(5,1), (5,2), (6,1), (6,2), (7,1), (7,2)}

{(5,1), (5,2), (6,1), (6,2), (7,1)}

B×A={(5,1),(5,2),(6,1),(6,2),(7,1),(7,2)}

To find the greatest common divisor (GCD) of 24 and 60, you can use various methods such as prime factorization or the Euclidean algorithm.

The correct option is: 12.

The distance between -3 and 11 is -----------_.Select the correct option

20

-31

1

14.

Consider the relation R = {(1, y), (1, z), (3, y)}, then R-1 is ______________.Select the correct option

empty set

None

{(1, y), (1, z), (3, y)}

{(y,1),(z,1),(y,3)}

{(y,1),(z,1),(y,3)}.

Suppose a and b are nonzero integers. Then gcd(a,b) x lcm(a,b)=|ab|.Select the correct option

Given statement is false

The given statement is true.

is true.

gcd(31, 15) = _____________.Select the correct option

none

31

1

15

gcd is 1.

Prime factorization of 90 is ___________.Select the correct option

3 × 15 × 2

3 × 3 × 5 × 2

9 × 5 × 2

none of these

Let A be a set of real numbers. If for every x ∈ A, |x| ≤ M, where M is a real number, then ___________.Select the correct option

A is bounded

A is unbounded

A is bounded.

The set Q of rational numbers is complete.

T

f

False.